CHEM1043 Tutorial 2

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Topic 2
Structure and Bonding – The Delocalized Approach

- Molecular Orbital Theory
- Homonuclear Diatomic molecules of 1st period
- Homonuclear Diatomic molecules of 2nd period
- Heteronuclear Diatomic molecules
- Delocalized Electrons: Bonding in Ozone
- Delocalized Electrons: Bonding in Benzene
- Band Theory

Molecular Orbital (MO) Theory

- Molecular orbital (MO) theory treats bonding in terms of molecular orbitals that can extend over an entire molecule

- Linear combination of atomic orbitals (LCAO): As two atoms approach each other, the wave functions of their atomic orbitals (AOs) combine to form a set of new molecular orbitals (MOs)

- The wave functions of electrons initially localized on different atoms begin to interact with one another and form new wave functions that represent chemical bonds in molecules (i.e. the new wave functions are delocalized)

LCAO

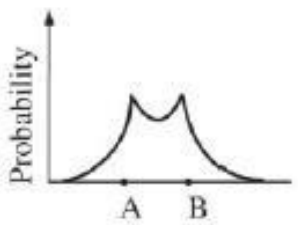
Constructive interference: bonding orbital

$$\Psi_1 (\sigma_{1s}) = \varphi_1 (1s_A) + \varphi_2 (1s_B)$$

The positive sign puts the waves in phase.



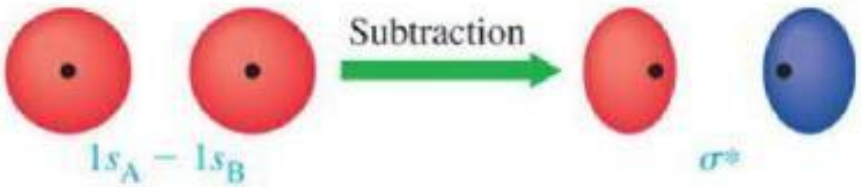
Greater electron probability between the nuclei: extra charge density between the nuclei: **bonding** molecular orbital



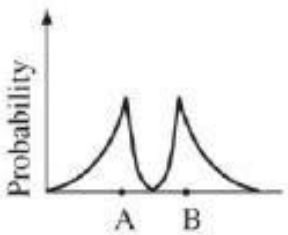
Destructive interference: antibonding orbital

$$\Psi_2 (\sigma^*_{1s}) = \varphi_1 (1s_A) - \varphi_2 (1s_B)$$

The negative sign puts the waves out of phase.



Lower electron charge density between the two nuclei in σ^*_{1s} orbital, strong repulsion occurs and the bond is weakened, hence the term "**antibonding**".



node

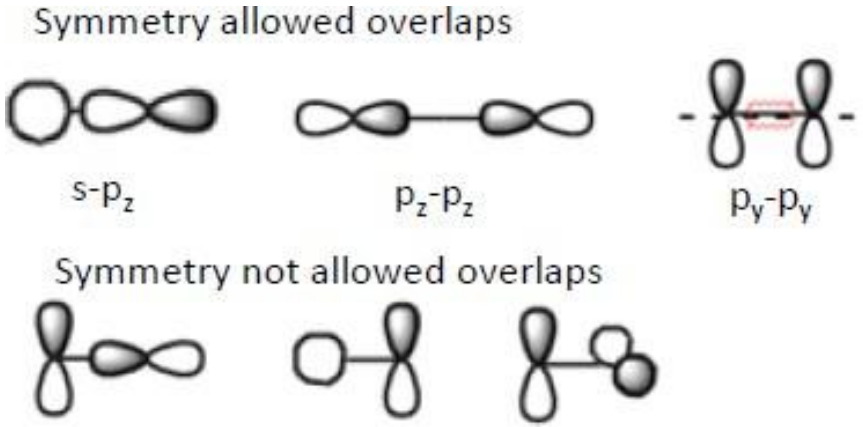
Basic Ideas of MOs

- General Principles:

1. No. of AOs combined = No. of MOs formed

2. Combination of two AOs forms two MOs, one bonding (lower energy) and one antibonding (higher energy)

3. Only AOs of similar energy and symmetry combine to form MOs



Basic Ideas of MOs

- Filling electrons:

4. Fill MOs from low to high energies (Aufbau Process)

5. Each MO can accommodate up to two electrons (Pauli Exclusion Principle)

6. Fill degenerate orbitals singly before pairing up (Hund's Rule)

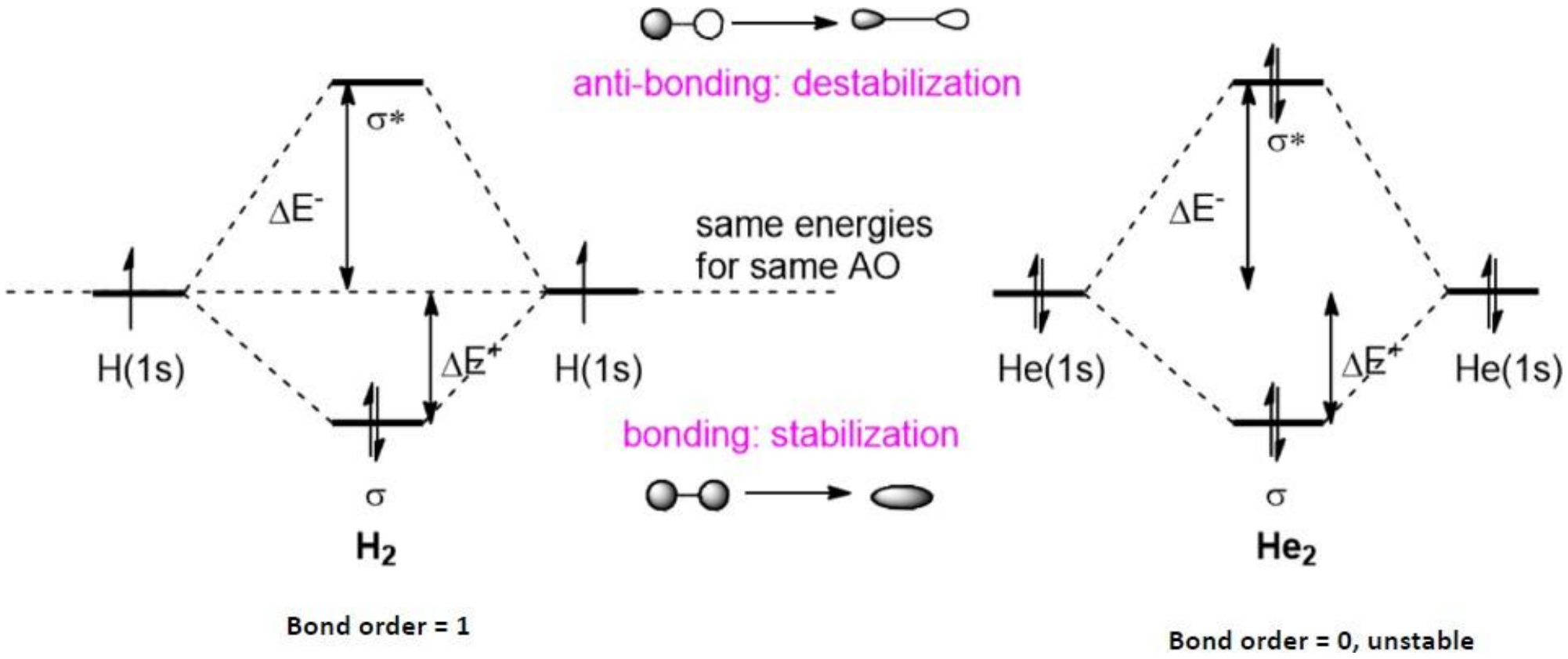
Basic Ideas of MOs

- Bond order:

7. Bond order = $\frac{no.of\ e^{-}in\ bonding\ MOs - no.of\ e^{-}in\ antibondingMOs}{2}$

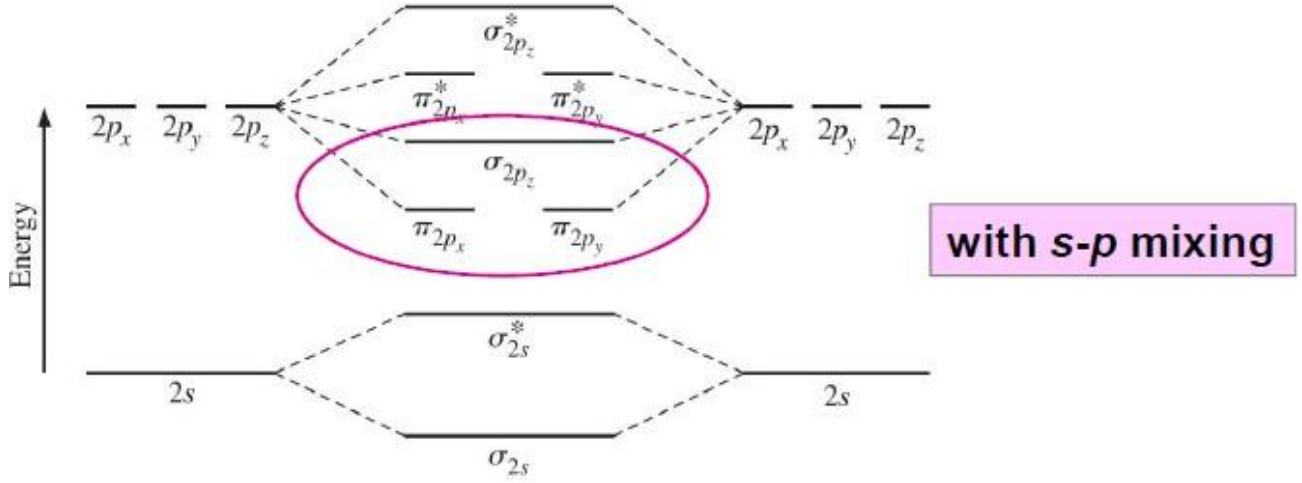
Homonuclear Diatomic Molecules

- 1St period



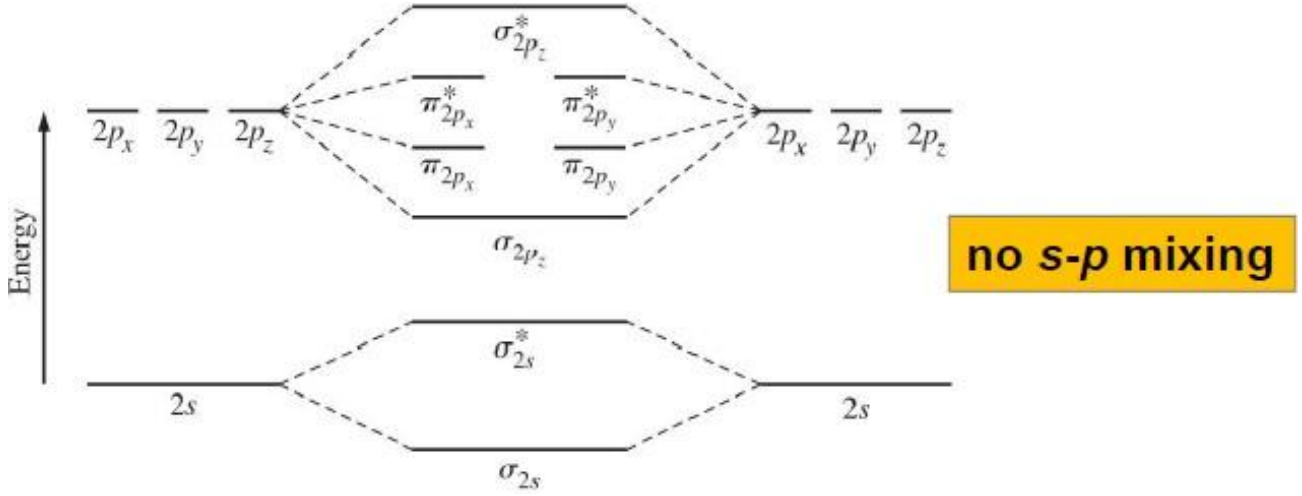
Homonuclear Diatomic Molecules

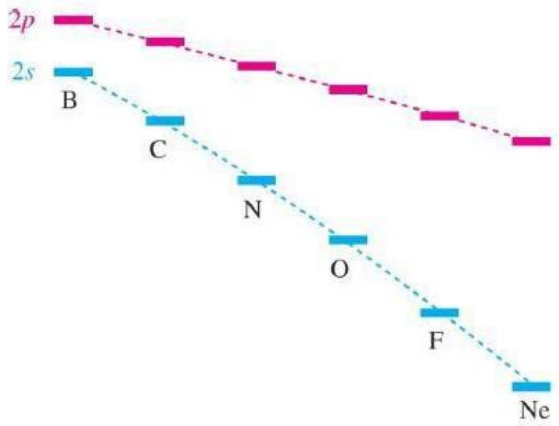
- 2nd period
- s-p mixing (z≤7)
- σ_{2s} and σ_{2p}: similar energy and shape → interact
- σ_{2p} is pushed up in energy above the π_{2p} orbitals



Homonuclear Diatomic Molecules

- 2nd period
- No s-p mixing (z≥8)
- s-p energy separation ↑, mixing effect ↓, σ_{2p} destabilization ↓

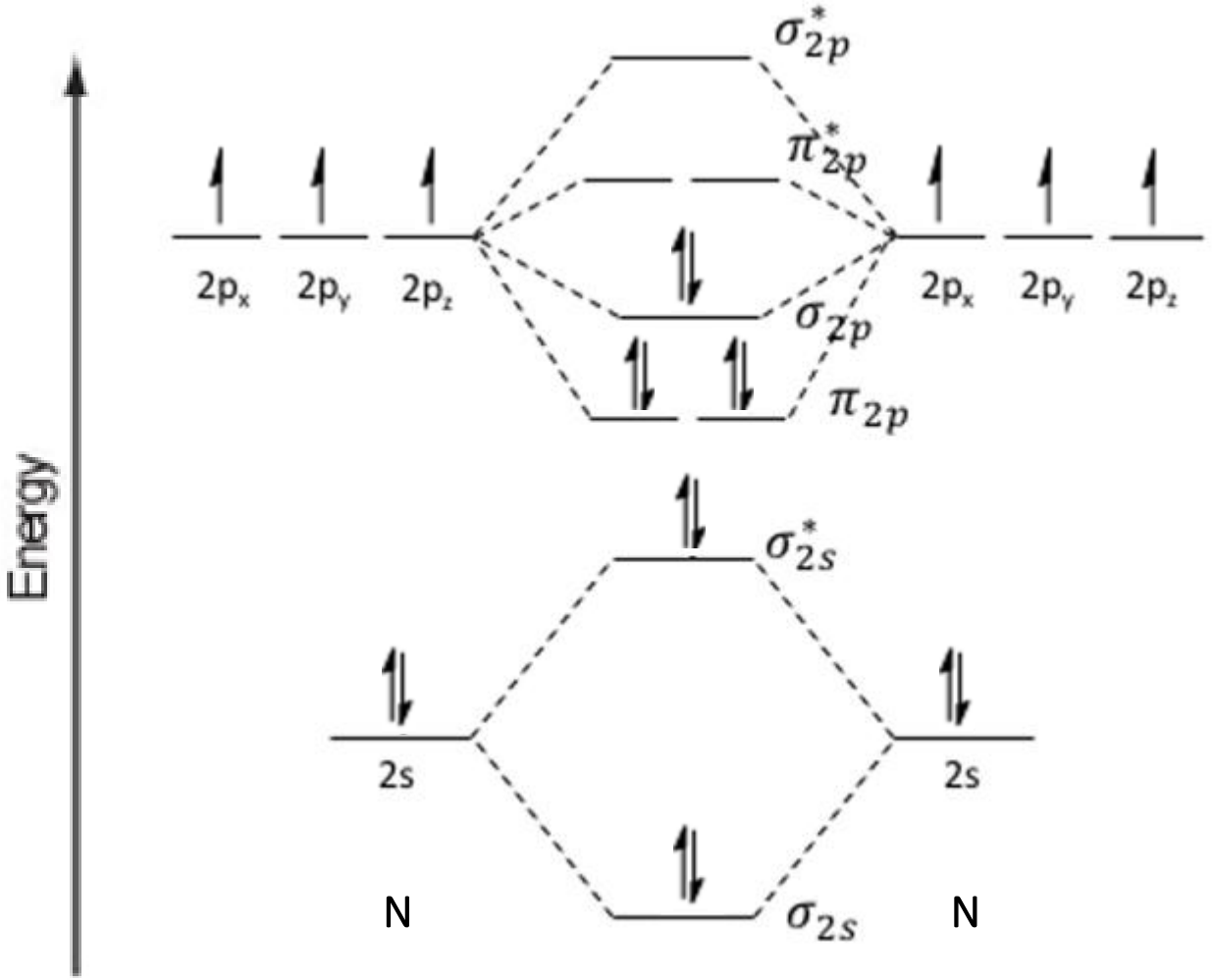




Question 1

- Draw the molecular orbital (MO) diagram for N_2.

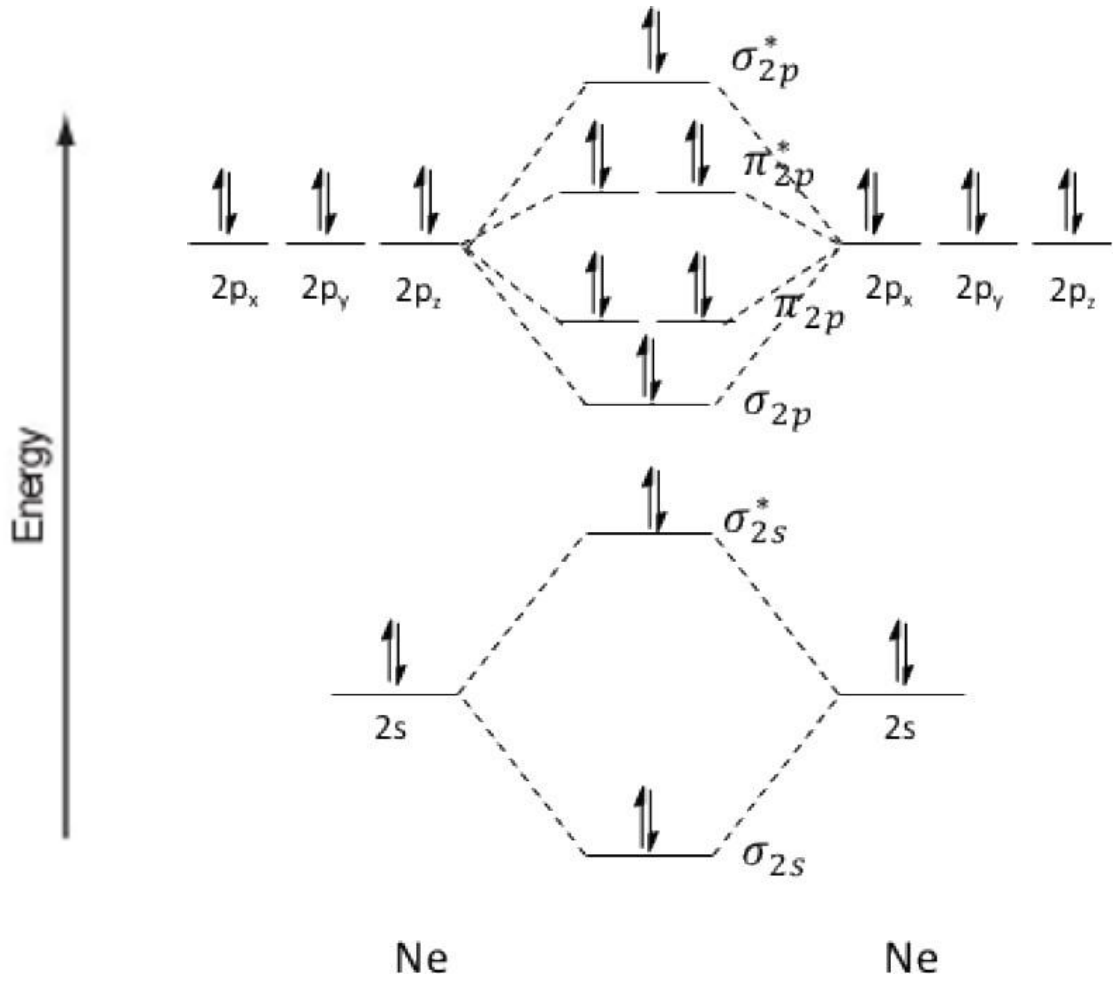
Question 1 answer



Question 2

- Use MO to explain why Ne does not tend to form diatomic molecule Ne_2.

Question 2 answer



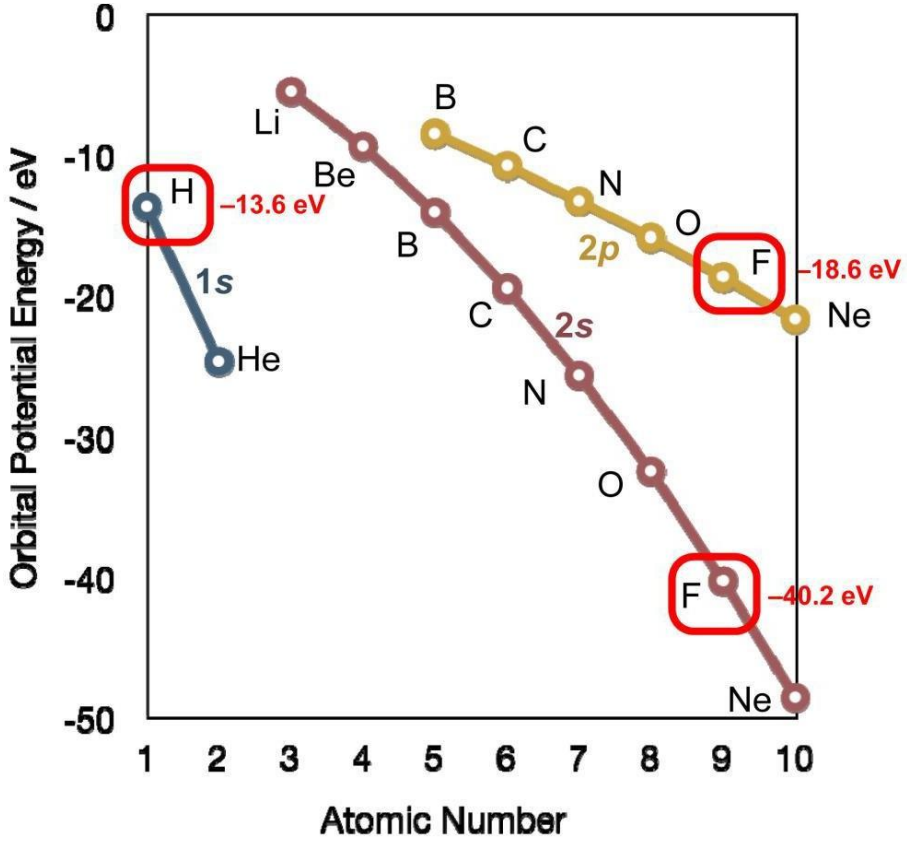
Bond order $= \frac{8-8}{2} = 0$

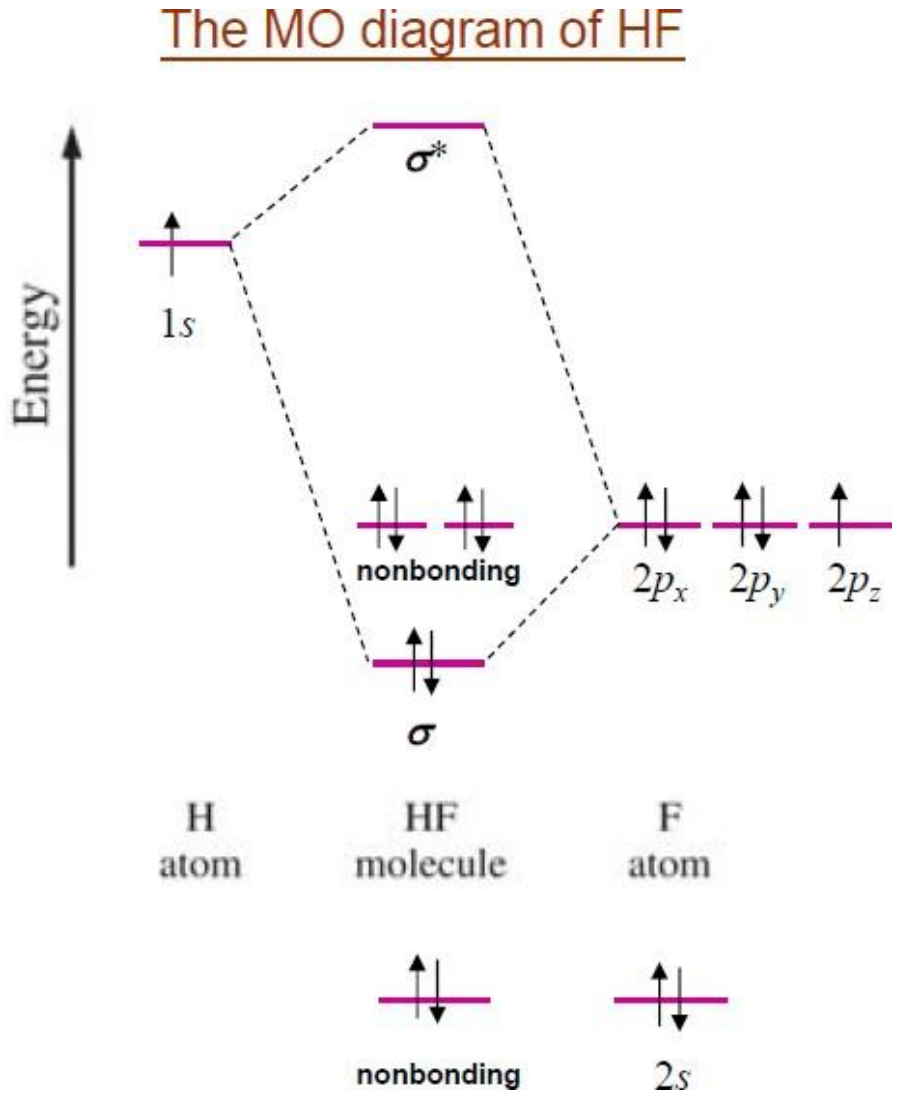
Unstable

→ does not form covalent compound

Heteronuclear Diatomic Molecules: HF

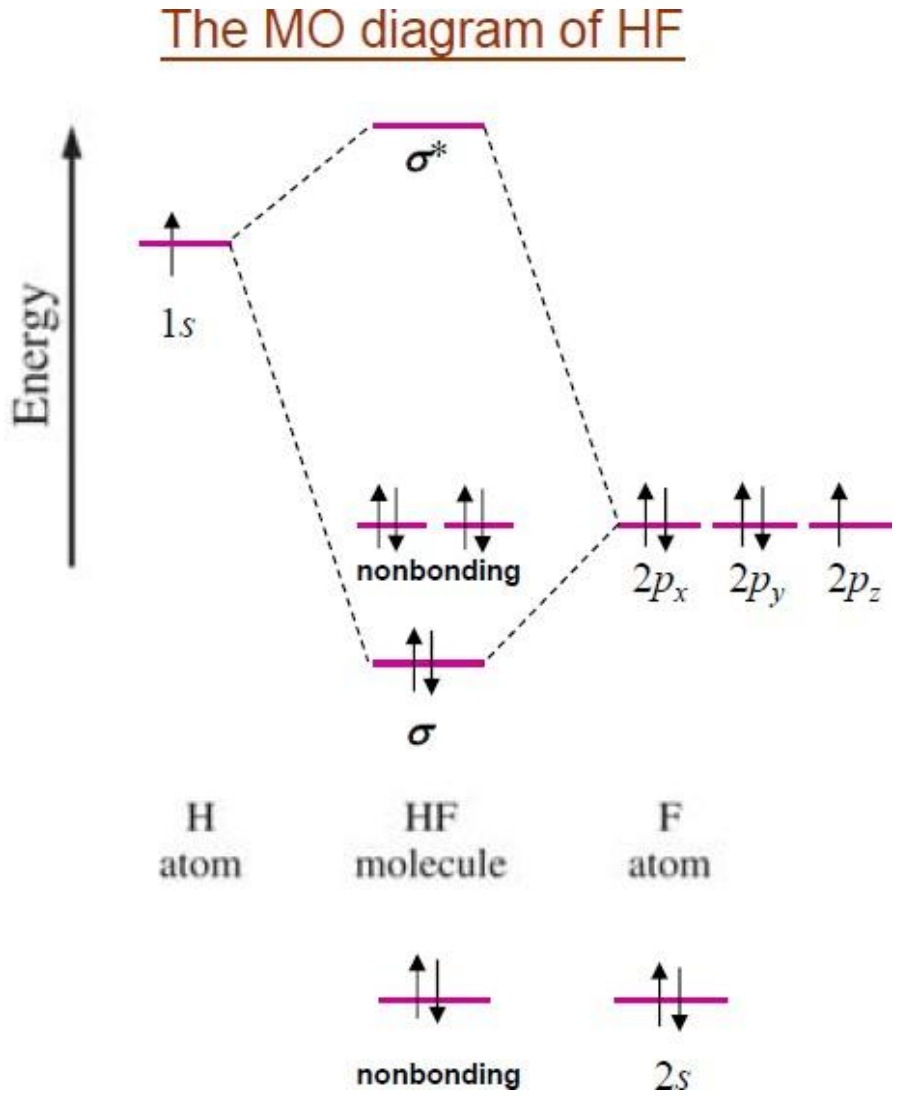
- F(2s) is deep in energy, no interaction expected





Heteronuclear Diatomic Molecules: HF

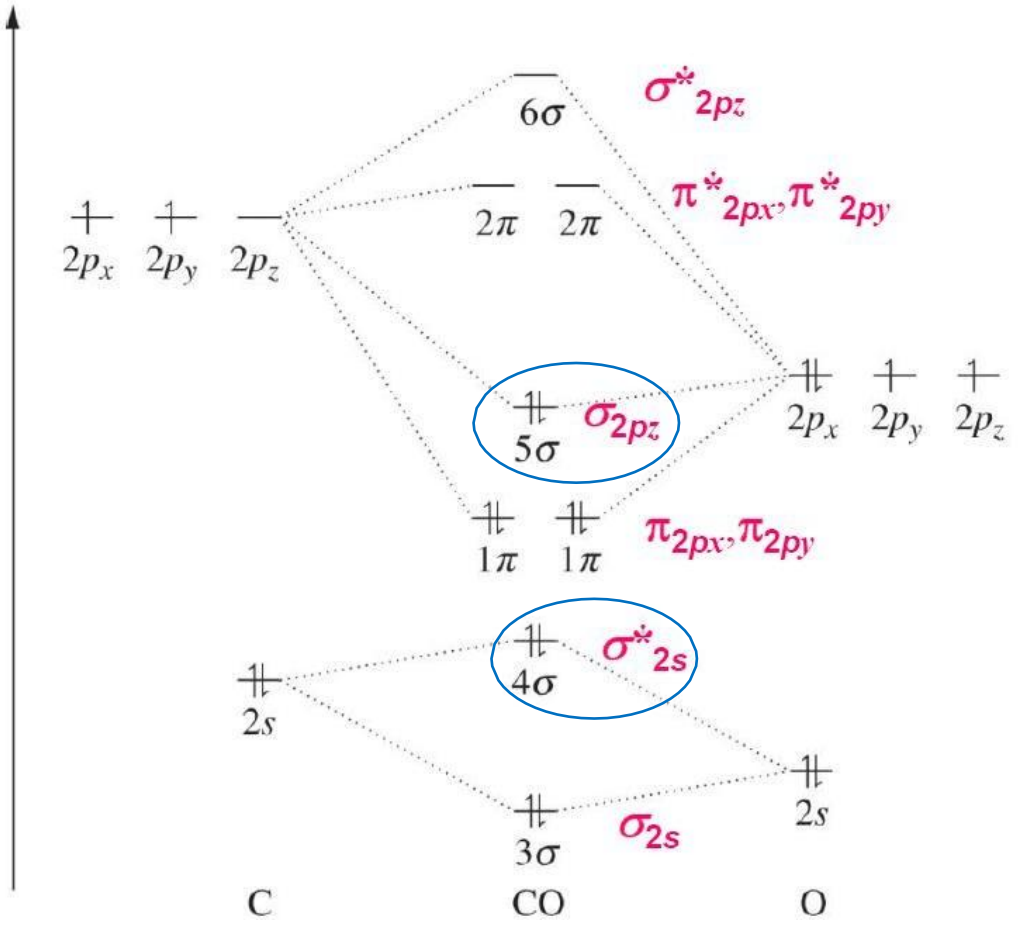
- Stable
- Both electrons are lowered in energy relative to their energy in the free H and F atoms



Heteronuclear Diatomic Molecules: CO

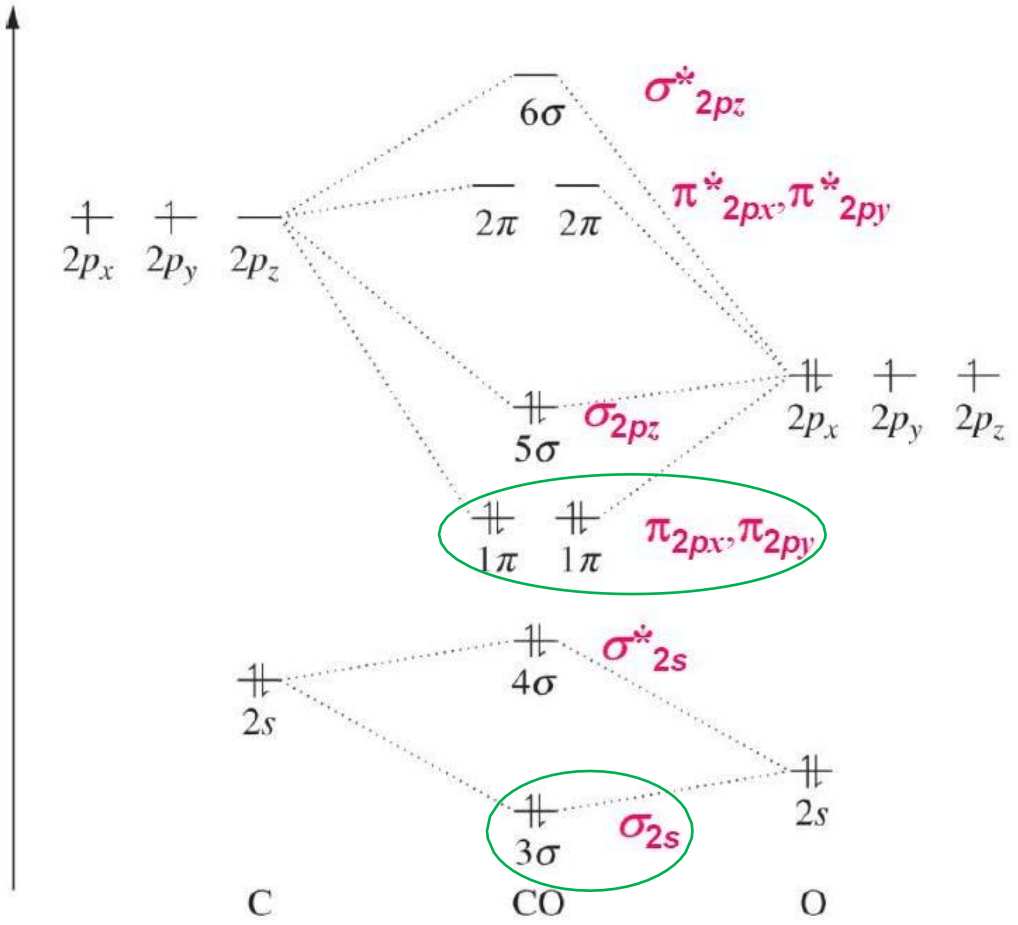
- 4σ – approximately halfway between 2s and 2p on O
- This means 4σ is mainly formed by mixing O 2s and O 2p. It is primarily O-based, so it is not strongly involved in C–O bonding. The 4σ can be classified as non bonding.

- 5σ – approximately halfway between 2s and 2p on C



Heteronuclear Diatomic Molecules: CO

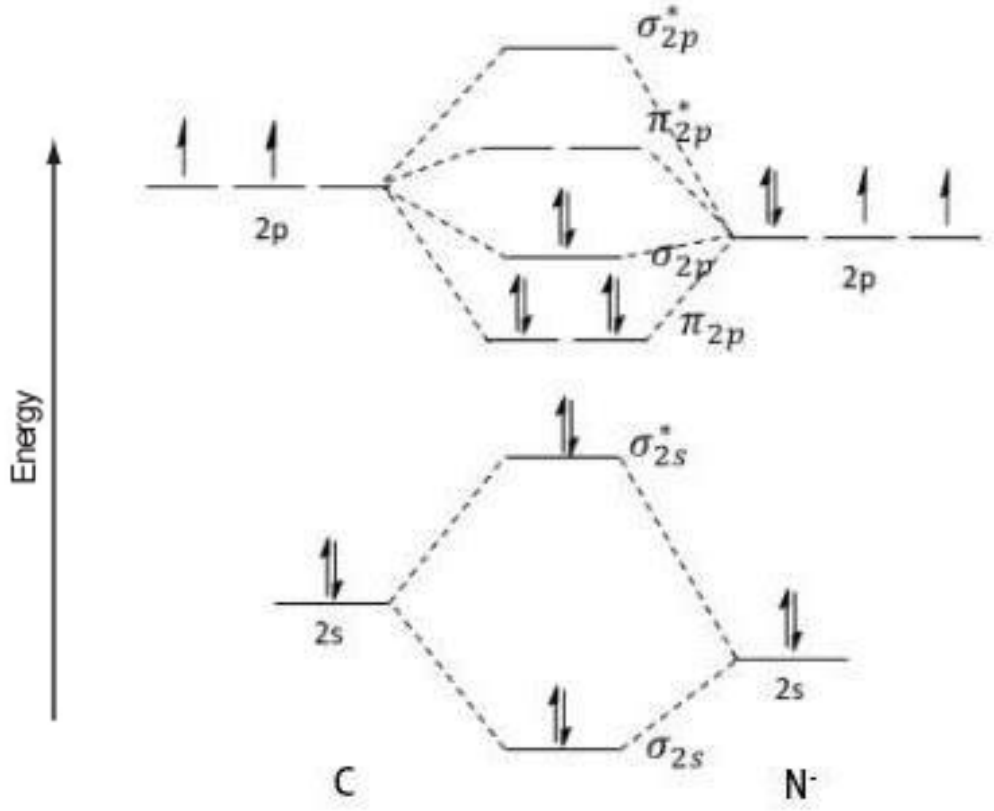
- Bonding of CO: 3σ and 1π
- Bond order = 6-0/2 = 3



Question 3

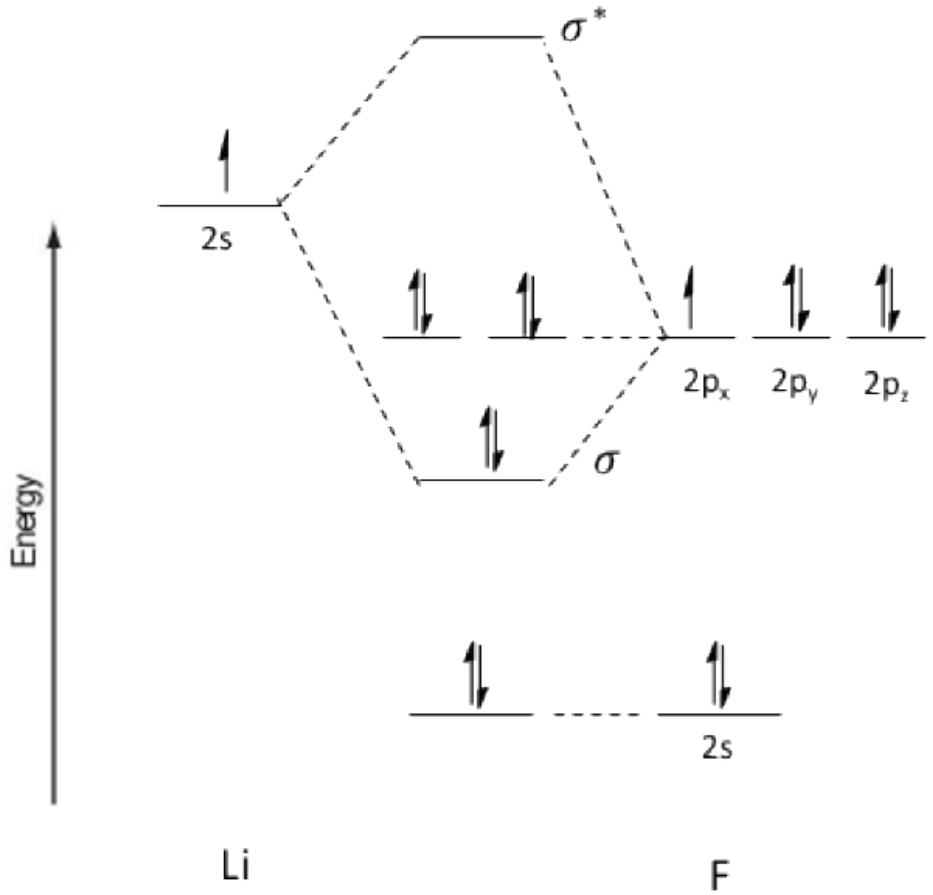
- Construct MO of CN^- and LiF. Predict the bond order and magnetic property.

Question 3 answer



Bond order $= \frac{6-0}{2} = 3$

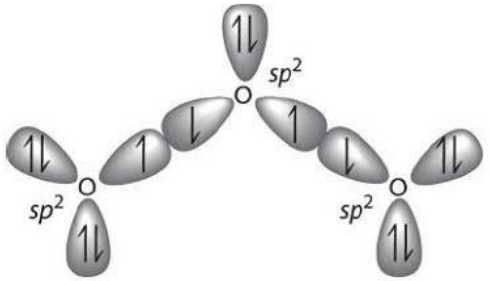
Diamagnetic



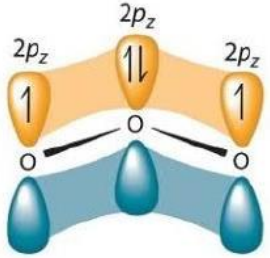
Bond order $= \frac{2-0}{2} = 1$

Diamagnetic

Delocalized Electrons: Bonding in Ozone

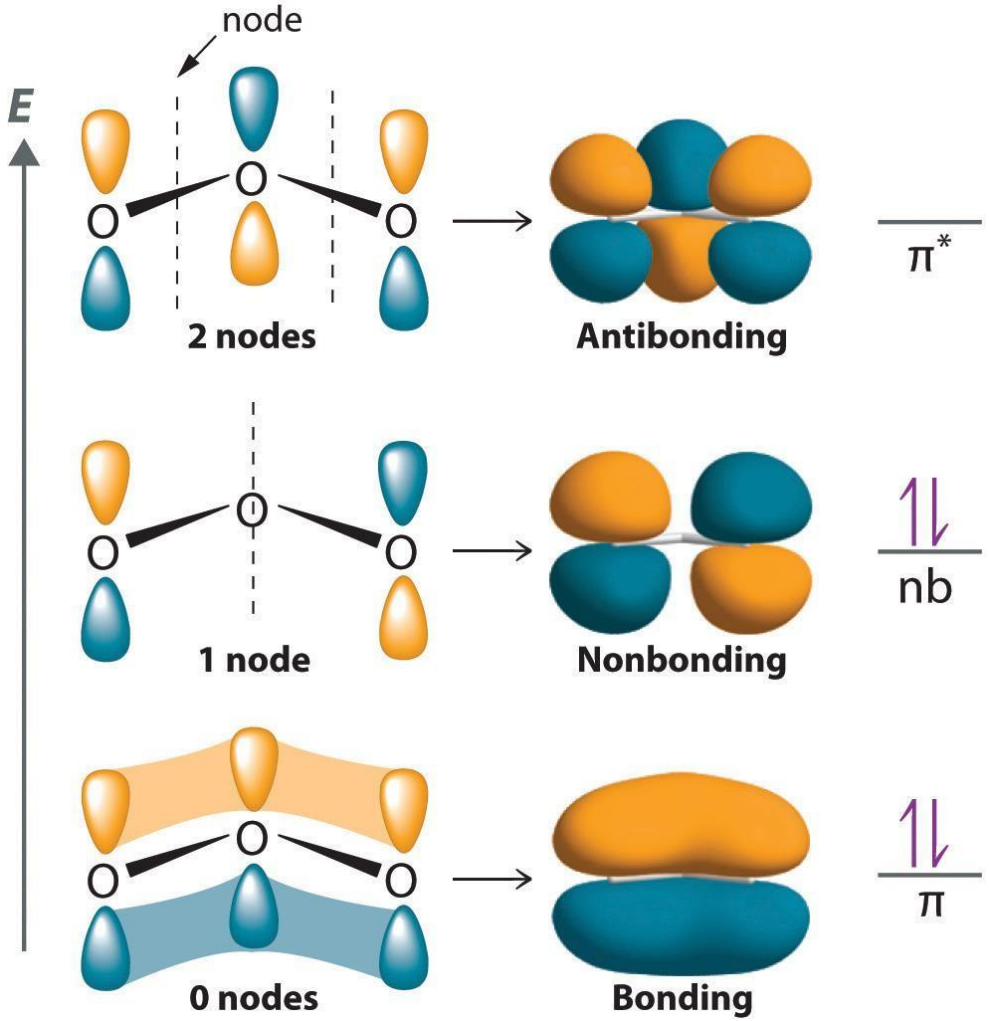


(a) σ bond framework



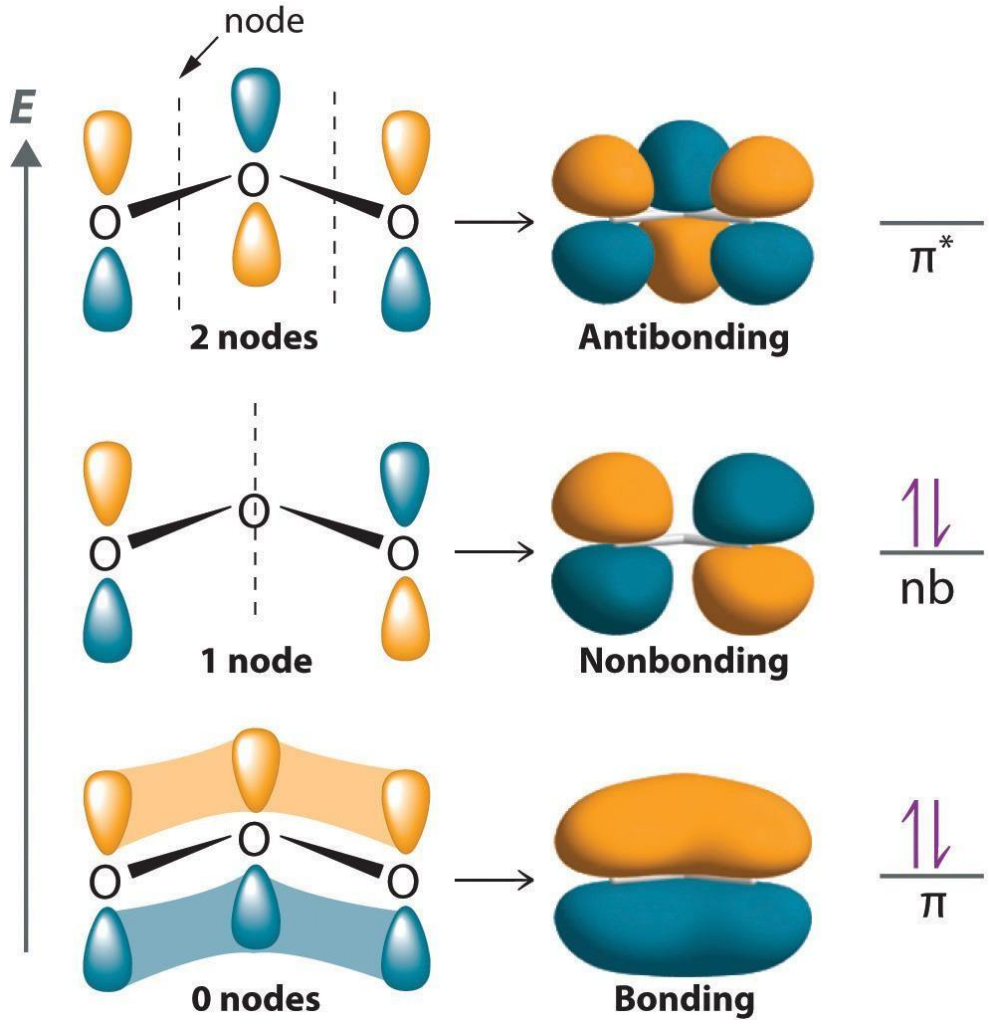
(b) Delocalized π molecular orbital

- Three 2p AOs of O atoms form three π MOs
  - 1 bonding, 1 non-bonding, 1 antibonding



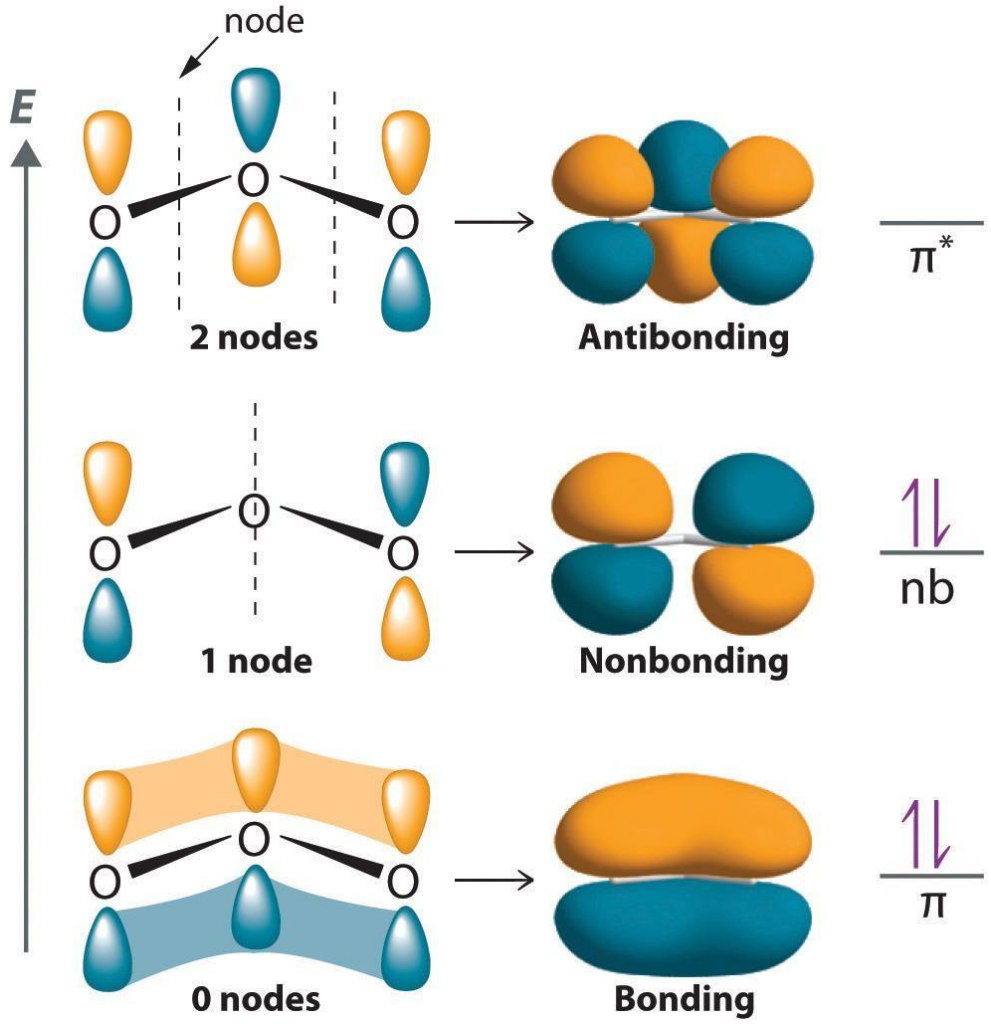
Delocalized Electrons: Bonding in Ozone

- Bond order of this π system
  $= \frac{2-0}{2} = 1$
- 1 bond is distributed among 2 O–O bonds
- Between each pair of O atoms
  $= \frac{1}{2} = 0.5$



Delocalized Electrons: Bonding in Ozone

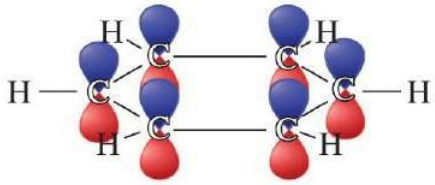
- Total bond order (σ+π) = 1 + 0.5 = 1.5 for each O–O bond



Delocalized Electrons: Bonding in Benzene

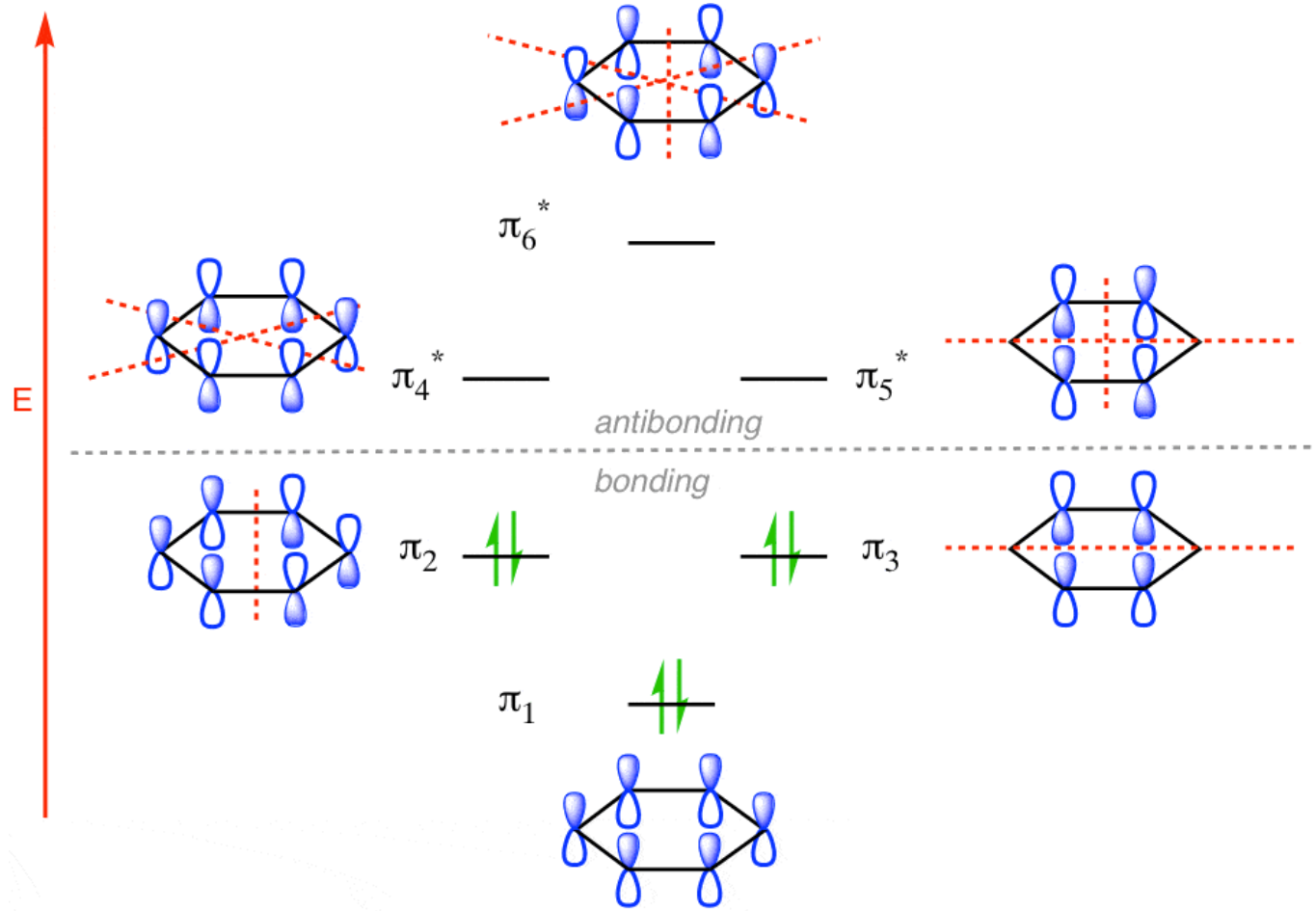


(a) σ bond framework



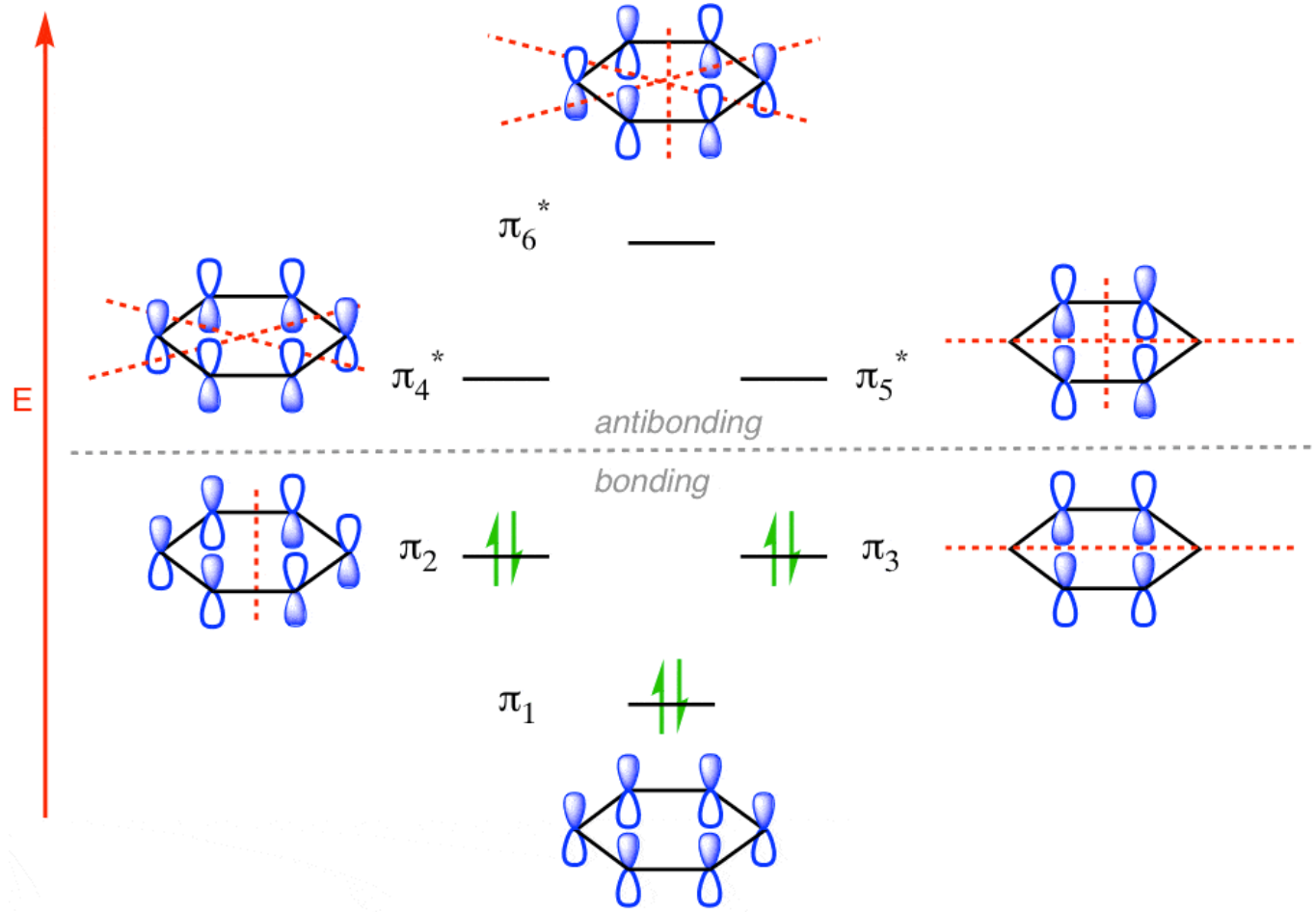
(b) Carbon 2*p* orbitals to be used in π bonding

- Six 2p AOs of C atoms form six π MOs
  - 3 bonding, 3 antibonding
- Energy of orbitals ↑, no. of nodes ↑



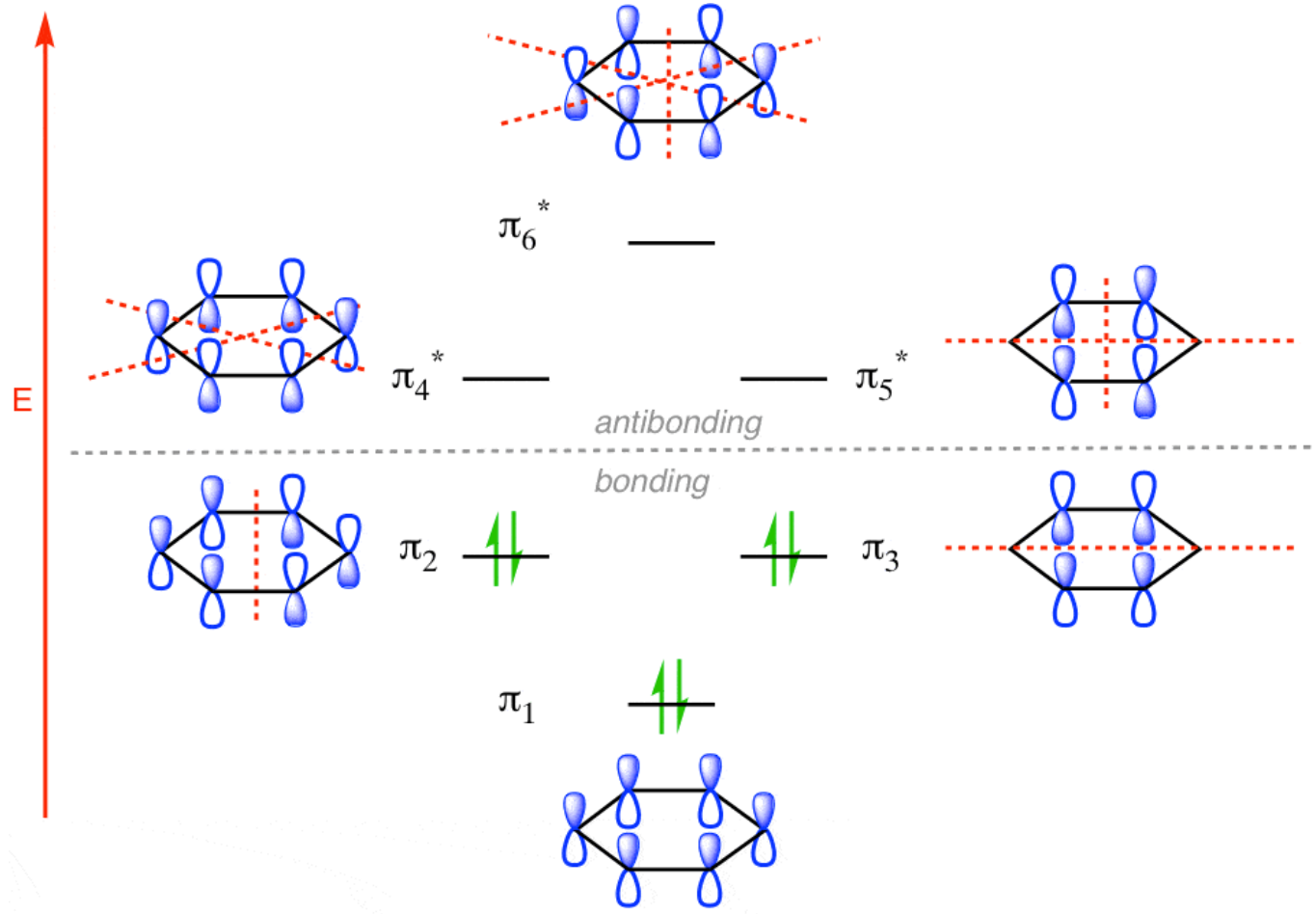
Delocalized Electrons: Bonding in Benzene

- Bond order of this π system = $\frac{6-0}{2} = 3$
- 3 bonds are distributed among six C atoms
- Between each pair of C atoms = $\frac{3}{6}$ = 0.5



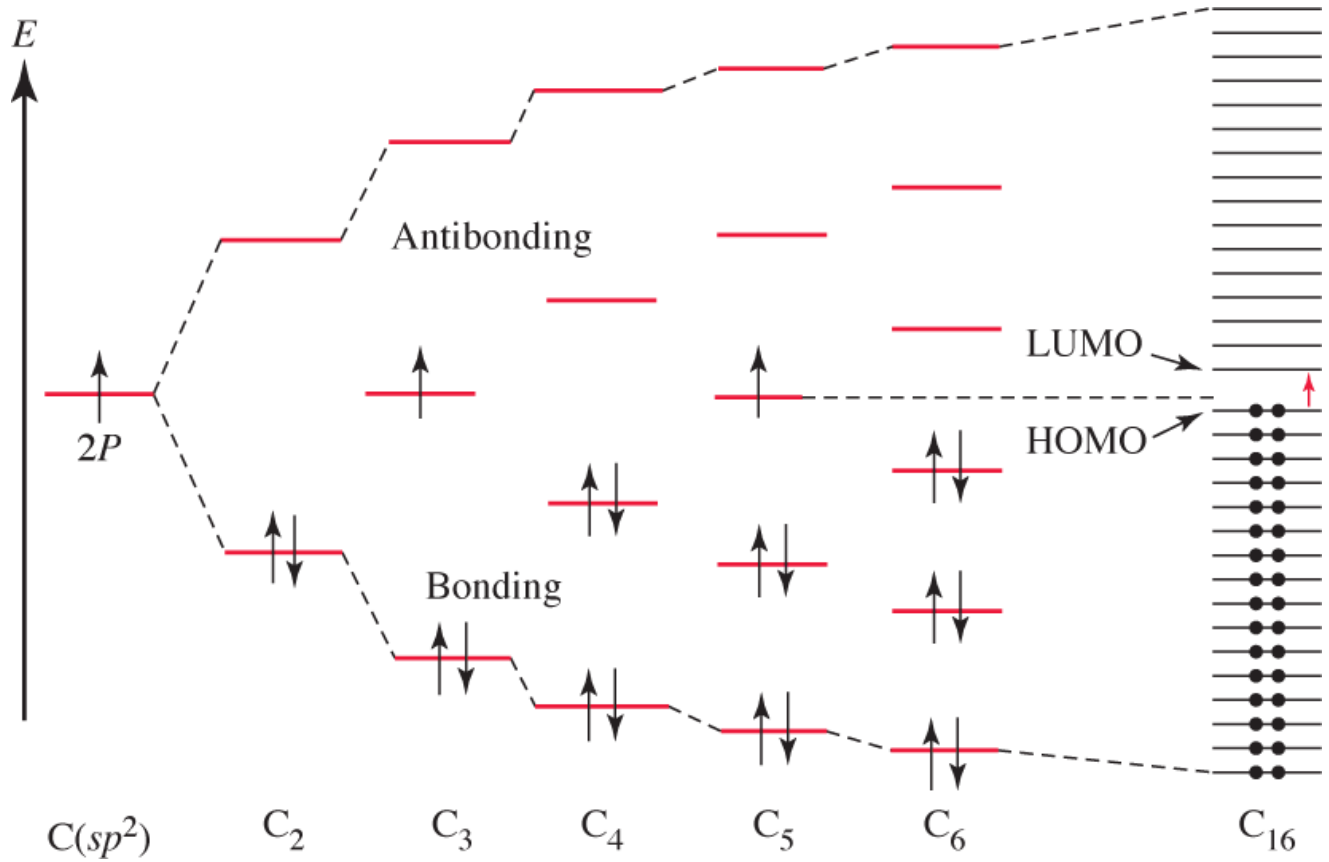
Delocalized Electrons: Bonding in Benzene

- Total bond order (σ+π) = 1 + 0.5 = 1.5 for each C–C bond

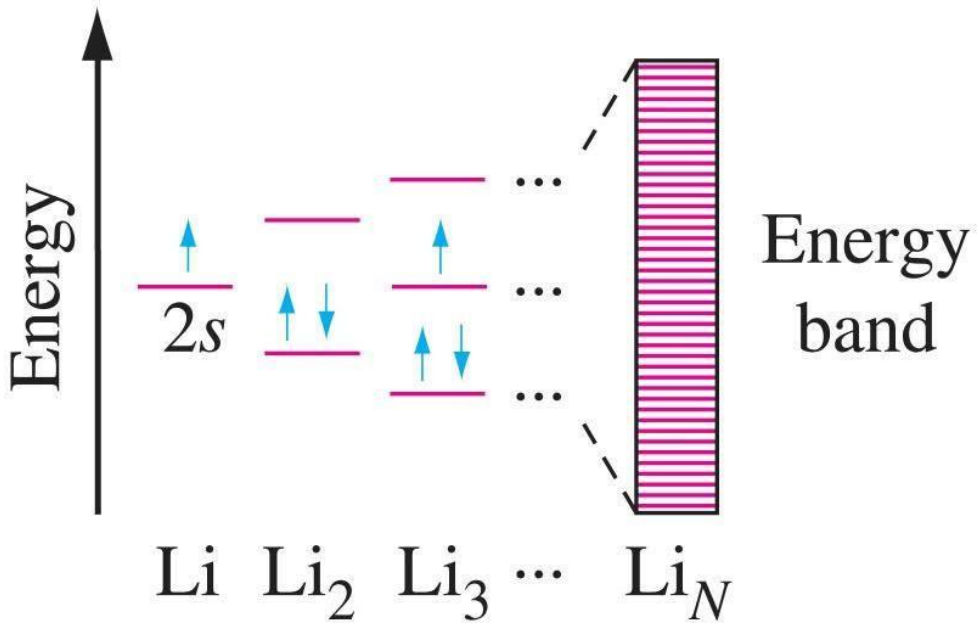


MO in Extended π Systems

- More extensive π system
- ↓
- Smaller energy gap between successive levels
- ↓
- Smaller HOMO-LUMO gap
- ↓
- Absorption at longer wavelength



Band Theory



- A set of *N* molecular orbitals with an extremely small energy separation between each pair of successive levels is resulted. This collection of very closely spaced molecular orbital energy levels is called an ***energy band***.

- Lower N/2 levels are filled with electrons (the valence band), and the upper N/2 levels are empty (the conduction band).

Band Theory: Electrical Conductivity



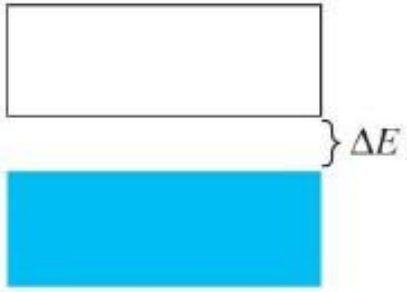
(a) Metal

- Partially filled valence band
- Also serves as the conduction band
- e.g. Na, Li
- When temp. ↑, conductivity ↓



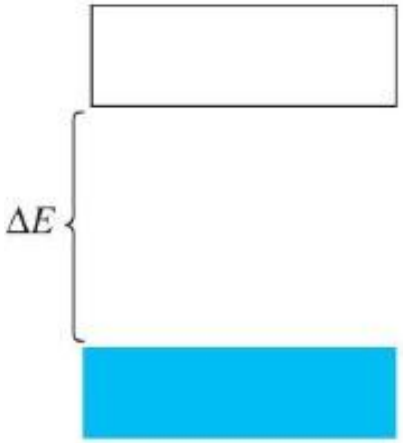
(b) Metal

- Full valence band
- Overlaps with the conduction band
- e.g. Be
- When temp. ↑, conductivity ↓



(c) Semiconductor

- Full valence band
- Empty conduction band
- Small energy gap
- e.g. Si, Ge
- When temp. ↑, conductivity ↑



(d) Insulator

- Full valence band
- Empty conduction band
- Large energy gap
- e.g. diamond, silica